

A modularity-matrix equivalent of the Fiedler value

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Reply to the meeting of Monday 31 August. Tested exclusively on Paper 1’s FAAMUNG data, as agreed.

Code in `modularity_indicator/`; every number below is printed by
`modularity_indicator/report_numbers.py`.

1 The question, and the answer in one page

Paper 1 (Cardoso & Palomar 2020) learns a graph Laplacian $\mathbf{L}_t$ directly from returns on rolling 30-day windows and reads one number off it: the Fiedler value $\lambda_2(\mathbf{L}_t)$, which measures how tightly the network holds together and is used as a market-stress indicator. The task set on 31 August was to find a quantity derived from the **modularity matrix** $\mathbf{B} = \mathbf{A} - \mathbf{d}\mathbf{d}^\top/2m$ that carries the same information, tested on Paper 1’s own data, as a first step toward a theory-backed account of what the spectra of these matrices encode.

Method. Paper 1 discards the learned graphs after extracting λ_2 , so Algorithm 2 was re-run with its exact constants to recover all **243** Laplacians, and the weighted adjacency recovered as $\mathbf{A}_t = -\text{offdiag}(\mathbf{L}_t)$. From each, 22 modularity-derived quantities were computed alongside 6 Laplacian references, and each candidate was scored on the four things Paper 1 actually uses λ_2 for: agreement with λ_2 , agreement of the trading gate, behaviour into the COVID crash, and the S1/S2 backtest. Two gates passed first: the recomputed λ_2 matches Paper 1’s cached series to 1.2×10^{-10} , and the backtest reproduces its published $S1 = +0.0779$ against $S2 = -0.0111$.

Result. The question as posed has a clean negative answer, in two parts.

The two-part answer

(i) **Normalise, and the question dissolves.** After degree normalisation the modularity matrix and the Laplacian are *the same operator*: they share an eigenbasis exactly, and their eigenvalues are in one-to-one correspondence $\mu = 1 - \lambda$. Verified across the full spectrum on all 243 windows to 1.8×10^{-15} . Any indicator built from normalised $\mathbf{B}$ is therefore a Laplacian indicator in different notation. This is a proof, not a measurement.

(ii) **Do not normalise, and the natural shortcut is unsound.** The candidate that topped our table, $\bar{d} - \mu_1(\mathbf{B})$, rests on an identity that holds only when *every* degree is equal. Substituting the *mean* degree is not a weakened hypothesis; it is no hypothesis. Its error is 89% as large as λ_2 ’s own standard deviation. This is the point raised at the meeting, and it is confirmed quantitatively in §4.

Three further findings came out of chasing that answer down, and they are worth more than the original question was.

- 1. Correlation, the metric we screened candidates with, is the wrong test for novelty.** It promoted the spurious candidate in (ii) to the top of the table, and it independently fails to flag a quantity that is provably pure tautology (§5).
- 2. Under Paper 1’s own experimental conditions, λ_2 sits inside the noise band.** It beats a random *day*, but not a random 24-day *stretch*: COVID lands at the 76th percentile of the sliding-event null, $p = 0.243$ (§6). Matching λ_2 was therefore never a well-defined target.
- 3. λ_2 splits into a structure part and a degree part, and the degree part is where $\mathbf{B}$ and $\mathbf{L}$ actually differ.** Degree information detects nothing on its own (AUC 0.537, a coin flip) yet *improves* detection when combined with structure, and it is exactly what normalisation destroys (§7). This localises where anything new can live, and reframes the project.

The natural first move is the one in (i): normalise both matrices and compare like with like. That is where the argument starts.

2 Degree normalisation, and why it collapses the question

What degree normalisation is

The **degree** $d_i = \sum_j A_{ij}$ is the total strength of stock i 's connections. The raw matrices $\mathbf{A}$ and $\mathbf{L}$ mix two different things: how *much* a node is connected (its size), and how that connection is *distributed* across its neighbours (its shape). A heavily connected stock dominates every spectral quantity simply because its numbers are larger.

Degree normalisation divides that out. Each entry becomes

$$A_{ij} \mapsto \frac{A_{ij}}{\sqrt{d_i d_j}},$$

so every edge is measured relative to how connected its two endpoints already are. A weight of 0.5 between two hubs is small; the same 0.5 between two peripheral stocks is large.

The analogy that makes this exact

Normalisation is to $\mathbf{L}$ what **correlation** is to **covariance**. Covariance mixes each series' volatility with their co-movement; dividing by $\sigma_i \sigma_j$ strips out the volatilities so only co-movement remains. Here, dividing by $\sqrt{d_i d_j}$ strips out the degrees so only structure remains. Same operation, same purpose.

Writing $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$, the two normalised objects are

$$\mathbf{L}_{\text{sym}} = \mathbf{D}^{-1/2} \mathbf{L} \mathbf{D}^{-1/2} = \mathbf{I} - \mathbf{N}, \quad \mathbf{B}_n = \mathbf{D}^{-1/2} \mathbf{B} \mathbf{D}^{-1/2}, \quad \mathbf{N} = \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2}.$$

The tell that the units are gone: doubling every edge weight doubles $\mathbf{L}$ but leaves $\mathbf{L}_{\text{sym}}$ untouched, and $\mathbf{L}_{\text{sym}}$'s eigenvalues always lie in $[0, 2]$.

The two normalised matrices are the same operator

Normalising the null term gives entries $\sqrt{d_i d_j}/2m$, which is exactly $\mathbf{u}\mathbf{u}^\top$ for the unit vector $\mathbf{u} = \mathbf{D}^{1/2} \mathbf{1} / \sqrt{2m}$, so

$$\boxed{\mathbf{B}_n = \mathbf{N} - \mathbf{u}\mathbf{u}^\top} \quad \text{and} \quad \mathbf{N}\mathbf{u} = \mathbf{u},$$

since $(\mathbf{N}\mathbf{u})_i = \sum_j \frac{A_{ij}}{\sqrt{d_i d_j}} \sqrt{\frac{d_j}{2m}} = \frac{d_i}{\sqrt{2m d_i}} = u_i$. So $\mathbf{u}$ is $\mathbf{N}$'s Perron eigenvector, and $\mathbf{B}_n$ does exactly one thing to $\mathbf{N}$: it deflates that one mode to zero and leaves every other eigenpair untouched. Hence

$$\text{spec}(\mathbf{B}_n) = \{0\} \cup \{1 - \lambda_i(\mathbf{L}_{\text{sym}})\}_{i \geq 2}, \quad \text{with the same eigenvectors.}$$

Equivalently $\mathbf{B}_n = \mathbf{I} - \mathbf{L}_{\text{sym}} - \mathbf{u}\mathbf{u}^\top$, where $\mathbf{u}$ is itself $\mathbf{L}_{\text{sym}}$'s null vector: $\mathbf{B}_n$ is a closed-form function of $\mathbf{L}_{\text{sym}}$ alone. Measured on all 243 windows:

Check (243 windows)	Result
$\max \text{spec}(\mathbf{B}_n) - (\{0\} \cup \{1 - \lambda_i(\mathbf{L}_{\text{sym}})\}) $	1.8×10^{-15}
$\max \mu_1(\mathbf{B}_n) - (1 - \lambda_2(\mathbf{L}_{\text{sym}})) $	1.3×10^{-15}
eigenvector agreement, $\mathbf{B}_n$ vs $\mathbf{L}_{\text{sym}}$ (mean best overlap)	1.000000 (min 1.000000)
for contrast: eigenvector agreement, $\mathbf{B}$ vs $\mathbf{L}$	0.871 (min 0.786)
for contrast: $ \langle \text{Fiedler vector of } \mathbf{L}, \text{ top eigenvector of } \mathbf{B} \rangle $	0.728 (range [0.039, 0.974])

Two things deserve emphasis. First, the pairing **reverses the order**: λ_2 is near the *bottom* of the Laplacian spectrum, and its twin μ_1 is at the *top* of the modularity spectrum. That is why the correspondence is not obvious by inspection, but it also means “read the top of $\mathbf{B}$ instead of

the bottom of $\mathbf{L}$ ” is the same idea backwards, not a new one. Second, the identity needs the graph **connected**, so that $\mathbf{L}_{\text{sym}}$ has a one-dimensional null space and $\mathbf{u}$ is recoverable. Paper 1 enforces this with $k = 1$, so it holds on all 243 windows here; it would fail on a fragmented graph, which is precisely the regime that broke the Paper 3 replication.

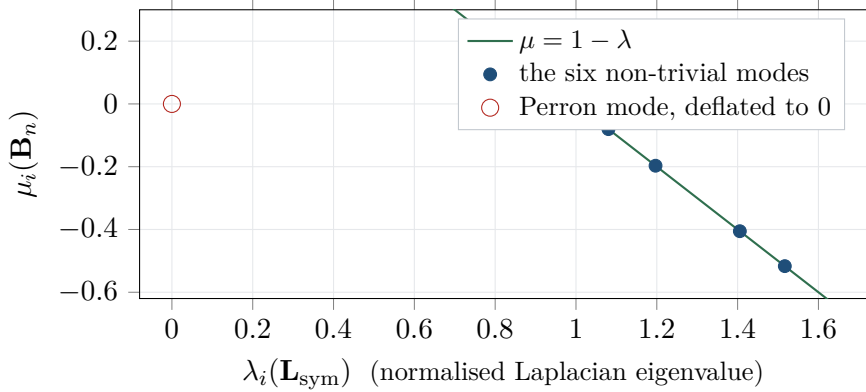


Figure 1: The spectra of $\mathbf{B}_n$ and $\mathbf{L}_{\text{sym}}$ for one window (2020-03-16, Paper 1’s own COVID-crash snapshot). Every eigenvalue lies on the line $\mu = 1 - \lambda$ exactly; the trivial mode is sent to 0 rather than to 1, which loses nothing because $\lambda_1(\mathbf{L}_{\text{sym}}) = 0$ always. The picture is identical on all 243 windows.

Consequence. No indicator derived from normalised $\mathbf{B}$ can be new. This closes a large part of the search permanently: not “we tried and failed” but “this cannot work”. So the search must use the *unnormalised* $\mathbf{B}$. Which raises the immediate question: how much of the menu we actually built was normalised?

3 The candidate menu, and how much of it was already decided

Of the 28 quantities scored, 22 are modularity-derived. Splitting them by provenance:

Built from	Candidates	Status
normalised $\mathbf{B}_n$ (3)	mu1_norm, muN_norm, one_minus_mu1n	tautological by §2
plain $\mathbf{B}$ (19)	mu1, mu2, muN, gap12, spread, mu_ratio, absmax, n_pos, trace_pos, trace_neg, energy, frob2, spec_entropy, q_bound, ipr1, iprN, v1_deg_align, dbar_minus_mu1, dual_resid	not settled by §2
Laplacian ref. (6)	lam2, lam_max, lam2_sym, dbar, two_m, deg_cv	baselines

The three normalised candidates are not merely correlated with Laplacian quantities; they are *affine images* of them, so their correlation is exactly ± 1 :

$$\text{corr}(\mu_1(\mathbf{B}_n), \lambda_2(\mathbf{L}_{\text{sym}})) = -1.0000000000, \quad \text{corr}(\mu_N(\mathbf{B}_n), \lambda_{\max}(\mathbf{L}_{\text{sym}})) = -1.0000000000,$$

and **one_minus_mu1n** equals $\lambda_2(\mathbf{L}_{\text{sym}})$ to *exactly* zero error. They were included as consistency checks and they behaved as such.

There is a second, less expected collapse *inside* the plain- $\mathbf{B}$ block. Several “different” candidates are one quantity relabelled:

Correlation with $\mu_1(\mathbf{B})$, 243 windows			
spread ($\mu_1 - \mu_N$)	+0.966	energy	+0.891
trace_pos ($\sum_{\mu>0} \mu$)	+0.951	mu_ratio	-0.881
gap12 ($\mu_1 - \mu_2$)	+0.935	q_bound	+0.793
frob2 ($\text{tr } \mathbf{B}^2$)	+0.898	muN	-0.668

So what looked like a table of five leading findings was largely $\mu_1(\mathbf{B})$ wearing different labels. (**absmax** is worse than a relabelling: since $|\mu_N| > \mu_1$ on 243/243 windows, it *is* $|\mu_N|$.) The one plain- $\mathbf{B}$ candidate that stands genuinely apart from the μ_1 family is $\mu_N(\mathbf{B})$, at -0.668 ; we return to it in §8.

One plain- $\mathbf{B}$ candidate topped the whole table on agreement with λ_2 : $\bar{d} - \mu_1(\mathbf{B})$, at 91% agreement of daily invest/cash calls. That is the one to examine next.

4 The theorem that was assumed, and why it does not hold

The motivation for $\bar{d} - \mu_1(\mathbf{B})$ is a genuine identity. On a d -**regular** graph, one in which every node has the *same* degree d , we have $2m = nd$, so $\mathbf{d}\mathbf{d}^\top/2m = (d/n)\mathbf{1}\mathbf{1}^\top$ and

$$\mathbf{B} = d\mathbf{I} - \mathbf{L} - \frac{d}{n}\mathbf{1}\mathbf{1}^\top \implies \mu_1(\mathbf{B}) = d - \lambda_2(\mathbf{L}).$$

The whole argument rests on $\mathbf{D} = d\mathbf{I}$: a scalar multiple of the identity commutes with everything and shifts every eigenvalue equally. For an irregular graph the term is $\mathbf{D}$, a diagonal matrix of *different* degrees, and there is no scalar to pull out. **Replacing d by the mean degree $\bar{d}$ is not a weakened hypothesis; it is no hypothesis at all**, and the resulting discrepancy is whatever it happens to be. This is the objection raised at the 31 August meeting, and it is why the $\lambda_2(\mathbf{L}) + \mu_1(\mathbf{B}) = d$ spine proposed in the 12 August log (direction D2) was not pursued. Measuring it:

Diagnostic	Value	Reading
degree CV of these graphs	0.305	nowhere near regular (0 would be); range [0.240, 0.369], $\bar{d} = 2.13$
$\max \text{spec}(\mathbf{B}) - (\{0\} \cup \{\bar{d} - \lambda_i(\mathbf{L})\}) $	1.25	$5.3 \times \lambda_2$'s own sd of 0.235. Not an identity
residual $r = (\bar{d} - \mu_1) - \lambda_2$, sd	0.208	89% of λ_2 's sd; the error is as large as the signal
residual, mean and range	0.492, [0.145, 0.971]	varies six-fold, so not a constant offset that could be subtracted
$\text{corr}(r, \lambda_2)$	+0.395	the error moves <i>with</i> the signal, inflating the apparent fit
$\text{corr}(r, \text{degree CV})$	+0.020	not a simple “more irregular $\Rightarrow$ bigger error” effect: no correction available
residual in-crash vs out	0.308 vs 0.512	the error is <i>event-dependent</i> ; part of the crash tracking is the error term

Decomposing the variance of the proxy itself, $\bar{d} - \mu_1 = \lambda_2 + r$: **only 40% comes from λ_2** , 32% from the residual, 28% from their covariance. Less than half of the movement in the proposed indicator is the quantity it was standing in for.

The $\bar{d}$ term was never doing any work. Controlling for graph scale, $\mu_1(\mathbf{B})$ on its own tracks λ_2 exactly as well as the proxy does:

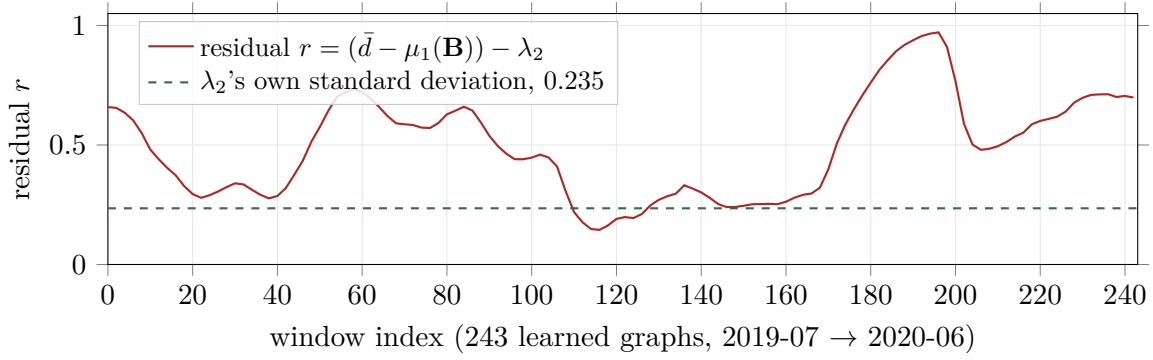


Figure 2: If $\bar{d} - \mu_1(\mathbf{B})$ were a usable proxy for λ_2 , this residual would be flat and small. It is neither: it swings from 0.15 to 0.97, with a standard deviation 89% as large as λ_2 's own. It is also systematically *lower* through the COVID window (indices 150–173), averaging 0.308 there against 0.512 outside, so part of what looked like crash tracking was the error term moving, not the identity holding.

	raw	partial, given $\bar{d}$
$\text{corr}(\lambda_2, \mu_1(\mathbf{B}))$	−0.574	− 0.851
$\text{corr}(\lambda_2, \bar{d} - \mu_1(\mathbf{B}))$	+0.856	+ 0.851

Identical. The subtraction only inflated the headline correlation by importing a scale nuisance; it contributed nothing conditional. The honest, defensible statement that survives is a regression fact with no theorem attached: *at fixed graph scale, $\mu_1(\mathbf{B})$ tracks λ_2 at −0.85.*

On the two correlation figures in the record

$\text{corr}(\lambda_2, \bar{d} - \mu_1)$ is 0.856 over all 243 learned graphs and 0.882 over the 201-day evaluation window. Both are correct; the earlier note quoted the second. All *detection* numbers in this report are on the 201-day window (that is where the backtest lives); all *structural* numbers are on all 243.

This leaves an uncomfortable question. A quantity with no valid derivation scored the highest agreement with λ_2 of anything we tested. Why did our screening let it through?

5 Why correlation is the wrong test for novelty

Correlation with λ_2 was adopted early as the working metric for comparing $\mathbf{B}$ -derived quantities against $\mathbf{L}$ -derived ones, and it is the first column of every table we produced. It is a reasonable measure of *agreement*. It is not a measure of *novelty*, and novelty is what the question was about.

What “novelty” means here: the definition this report commits to

A modularity-derived quantity is **new** if and only if it **cannot be reconstructed from** $\text{spec}(\mathbf{L})$. If it can, then whatever it measures was already available on the Laplacian side, and calling it a modularity indicator is a relabelling.

Necessary caveat. $\mathbf{L}$ determines $\mathbf{A}$ determines $\mathbf{B}$, so at the level of *matrices* everything here is recoverable from everything else and the question is empty. The claim is strictly about **spectra**, which is where indicators actually live: $\text{spec}(\mathbf{B}_n)$ is a bijection of $\text{spec}(\mathbf{L}_{\text{sym}})$ (§2), whereas $\text{spec}(\mathbf{B})$ is not a bijection of $\text{spec}(\mathbf{L})$, and they do not even share an eigenbasis (overlap 0.871, and on some windows the two leading modes are nearly orthogonal at 0.039).

Correlation fails against that definition in both directions.

Direction 1: it accepts things that are pure tautology. Correlation only detects an *affine*

relationship. When the map from a Laplacian quantity to a modularity quantity is affine, correlation catches it perfectly; when the map is non-linear, correlation goes quiet even though the quantity contains literally no new information.

Quantity of $\mathbf{B}_n$	corr. with λ_2	error when rebuilt from $\text{spec}(\mathbf{L}_{\text{sym}})$ alone
$\mu_1(\mathbf{B}_n)$ (affine: $1 - \lambda_2$)	-1.0000000000	n/a (exact by construction)
nuclear norm $\ \mathbf{B}_n\ _*$	-0.834	2.7×10^{-15}
spectral entropy of $\mathbf{B}_n$	-0.264	8.0×10^{-15}

The last row is the whole point. Spectral entropy of $\mathbf{B}_n$ correlates only -0.264 with λ_2 . On a correlation screen it looks like an independent signal and would have been promoted as a discovery. It was rebuilt here from the $\mathbf{L}_{\text{sym}}$ spectrum alone, with no access to $\mathbf{B}$ at any stage, to 8.0×10^{-15} . It carries exactly zero new information.

Direction 2: it promotes things that are spurious. That is §4: the candidate with the single highest correlation in the study, 0.882, had no valid derivation, and 40% of its variance was the target while 32% was error.

The practical consequence. *Low correlation with λ_2 is not evidence of new information, and high correlation is not evidence of a sound relationship.* λ_2 is one scalar summary of a spectrum; a non-linear re-reading of that same spectrum will decorrelate from it while adding nothing. Novelty has to be argued from what a quantity is a *function of*, not from how it correlates, which is why the criterion in the box above is stated in terms of reconstructibility.

Both directions of the test, though, measure candidates against λ_2 . That presumes λ_2 is a sound target. It is worth checking whether it is.

6 λ_2 under Paper 1’s own experimental conditions

The setup, stated precisely

universe	7 stocks (META, AAPL, AMZN, MSFT, UBER, NFLX, GOOGL)
sample	272 trading days, 2019-06-04 $\rightarrow$ 2020-06-30
graphs	243 causal rolling windows, 30 days each; $k = 1$, $\eta = 0$, $\beta = 0$, $\delta = 100$
evaluation window	201 signal days, 2019-07-16 $\rightarrow$ 2020-05-01 (exp4’s window)
events	one: COVID, 2020-02-19 $\rightarrow$ 2020-03-23, giving 24 crash days
rule	invest when $\lambda_2 < \tau = 1.0$; that gate invests 79/201 days (39.3%)
outcome	$S1$ (buy & hold) = +0.0779; $S2$ (λ_2 gate) = -0.0111

Seven stocks and a single event. That is the entire evidential base for the indicator, and it bounds what any comparison against it can establish.

λ_2 sits inside the noise band

- The raw indicator scores below chance.** At raw levels, λ_2 ’s COVID AUC is **0.472**. FAA-MUNG’s autumn-2019 regime is already elevated, so quiet days outrank crash days. Only after applying a strictly-prior 60-day trailing z -score does it reach 0.759. The published indicator, used as published, does not separate the crash.
- It beats a random day, but not a random stretch.** These are different nulls and they disagree:

Null	Result	Verdict
bootstrap over days	AUC 0.759, 95% CI [0.634, 0.875]	excludes 0.5: passes
sliding 24-day event window	COVID at the 76th percentile , $p = \mathbf{0.243}$ null median 0.434, 95th pct. 0.925, max 0.963	fails

λ_2 does rank crash days above calm days. What it cannot do is distinguish COVID from an arbitrary volatile 24-day stretch of the same series, which is what a crash *detector* is supposed to do.

- c. **The trading result is consistent with luck.** A uniform random gate at the same invest rate beats buy-and-hold **39.2%** of the time, so across 28 candidates roughly 11 were expected to beat $S1$ by chance alone; 5 did. A circular-shift null preserving each signal’s own autocorrelation clears nothing: the best-performing candidate, $\text{tr}(\mathbf{B}^2)$ at $S2 = +0.127$, has $p = 0.305$, and the smallest p -value achieved by any candidate is 0.115.
- d. **No multiplicity correction exists anywhere in this study.** 28 quantities were scored against a single event. Nothing in this report, ours or Paper 1’s, should be read as a corrected significance claim.

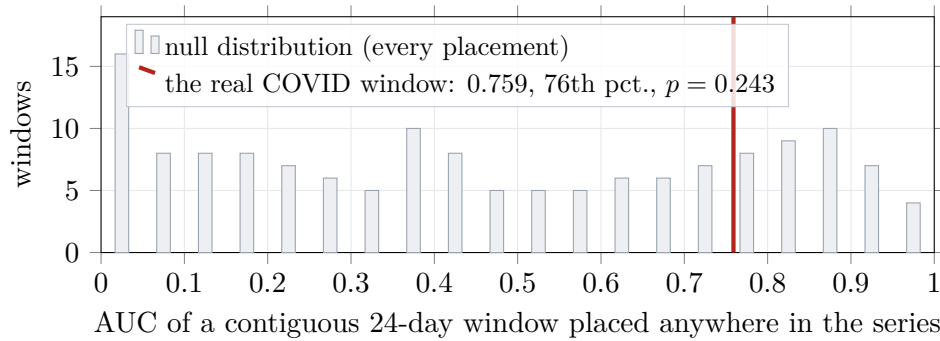


Figure 3: The sliding-event null for λ_2 ’s z -scored detector. If λ_2 were specific to COVID, the red line would sit in the far right tail. It sits inside the bulk: roughly a quarter of arbitrary 24-day windows score at least as well.

Consequence for the original question. Matching λ_2 was never a well-posed target, because λ_2 ’s own status on this data is not established. The productive move is not to find something that agrees with it, but to ask what it is *made of*.

7 Decomposing λ_2 into structure and degree

How the split is defined

The key observation is that Paper 1’s indicator is read off the **unnormalised $\mathbf{L}$** . By §2, normalisation removes exactly two things: overall scale, and the spread of degrees across nodes. So $\lambda_2(\mathbf{L})$ is not a pure structure measure. It is a blend of:

- a **structure** part, isolated by $\lambda_2(\mathbf{L}_{\text{sym}})$, which is scale-free by construction; and
- a **scale/degree** part, carried by $\bar{d}$ (equivalently $2m$) and by the degree spread deg_cv .

Two ways to check that this split is real and not an artefact of the fit:

		all 243 graphs	201-day window
$\text{corr}(\lambda_2(\mathbf{L}), \lambda_2(\mathbf{L}_{\text{sym}}))$	structure	+0.784	+0.887
$\text{corr}(\lambda_2(\mathbf{L}), \bar{d})$	scale	+0.489	+0.743
R^2 of $\lambda_2(\mathbf{L}) \sim \lambda_2(\mathbf{L}_{\text{sym}}) + \bar{d}$		0.837	0.848

A regression alone would be weak evidence. The containment is exact, however:

$$d_{\min} \cdot \lambda_2(\mathbf{L}_{\text{sym}}) \leq \lambda_2(\mathbf{L}) \leq d_{\max} \cdot \lambda_2(\mathbf{L}_{\text{sym}})$$

holds on 243/243 windows, with λ_2 sitting 0.24 of the way across a band of mean width 1.351 (sd of that position, 0.06). So “ $\lambda_2 = \text{structure} \times \text{scale}$ ” is a structural statement, not a fitted one: the structure term sets the value up to a factor bounded by the degree range.

What each part detects

Each component was put through the same detector (strictly-prior 60-day z -score, direction fixed by correlation with λ_2 , never by what scores best) and scored on the same 24 COVID days:

Quantity	What it is	AUC (z)	AUC (raw)
$\lambda_2(\mathbf{L})$	the full indicator, unnormalised	0.759	0.472
$\lambda_2(\mathbf{L}_{\text{sym}})$	structure only	0.700	0.594
$\bar{d}$	scale only	0.537	0.156
$2m$	total edge weight	0.537	0.156
deg_cv	degree spread	0.606	0.701
$\mu_N(\mathbf{B})$	plain modularity, most negative mode	0.775	0.791

Read the first three rows together, because that is the finding:

Degree information is a modifier, not a signal

Scale on its own is a coin flip (0.537). Structure on its own reaches 0.700. But the two *combined*, which is what unnormalised λ_2 is, reach 0.759, better than either part. The degree information carries no signal by itself, yet it improves the signal when mixed with structure.

And degree information is precisely what normalisation destroys. That is the mechanism behind §2: the normalised branch is dead not because normalising is wrong, but because it discards the one ingredient that was adding value.

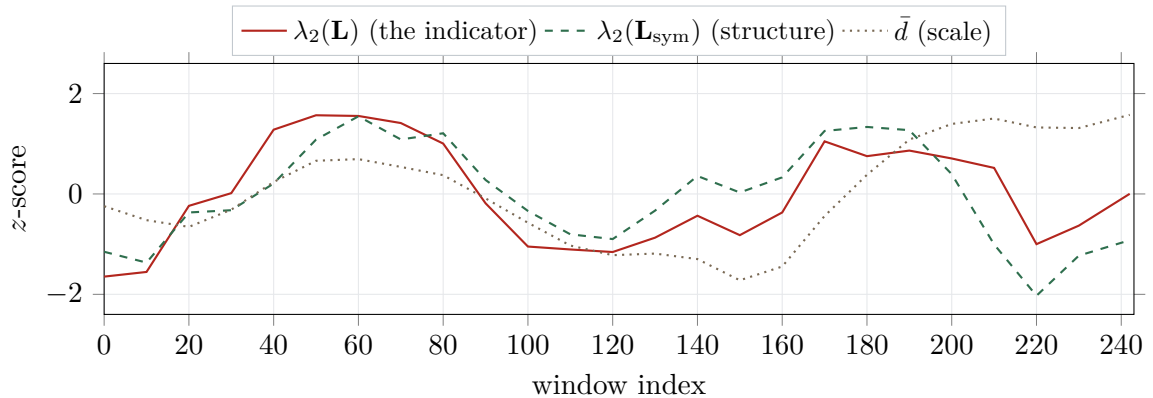


Figure 4: The indicator and its two ingredients, each standardised. The COVID window is indices 150–173. From index ≈ 190 (April 2020) the two ingredients pull apart, structure relaxing sharply while scale keeps rising, and the unnormalised λ_2 , being a blend, stays elevated on scale alone. That is the stretch in which Paper 1’s strategy sits in cash through the recovery and ends at -0.011 against buy-and-hold’s $+0.078$.

Why this matters

The degree sequence has to be combined with the structure somehow, and $\mathbf{L}$ and $\mathbf{B}$ use **different combination rules**:

$\mathbf{L} = \mathbf{D} - \mathbf{A}$	degrees enter additively , on the diagonal: “total connection, offset by structure”.
$\mathbf{B} = \mathbf{A} - \mathbf{d}\mathbf{d}^\top/2m$	degrees enter as a rank-one null model : “does the observed structure exceed what these degrees alone would predict, if edges were rewired at random?”

That second question is genuinely different, and by §2 it is the *only* thing $\mathbf{B}$ offers that $\mathbf{L}$ does not. It also explains, rather than merely observing, why $\mu_N(\mathbf{B})$ keeps scoring well: it sits at the end of the unnormalised modularity spectrum where the degree-dependent departure from the normalised limit is largest.

8 Final remarks and where to go next

What was established

The question as posed (find a modularity quantity carrying the same information as the Fiedler value) has a negative answer, and the negative answer is the deliverable. It *closes* the direction rather than leaving it to be re-attempted:

Route	Status	Why
normalised $\mathbf{B}$	closed by proof	$\mathbf{B}_n = \mathbf{I} - \mathbf{L}_{\text{sym}} - \mathbf{u}\mathbf{u}^\top$; same spectrum, same eigenvectors
μ_1 with $\bar{d}$ substituted	closed as unsound	needs $\mathbf{D} = d\mathbf{I}$; error is 89% of the signal’s sd
plain $\mathbf{B}$, μ_1 family	largely redundant	$ r \geq 0.79$ with μ_1 for six of the leading candidates
plain $\mathbf{B}$, negative end	open	μ_N at -0.668 with μ_1 ; no identity ties it to λ_2

The one surviving lead

$\mu_N(\mathbf{B})$, the most negative modularity eigenvalue (the “anti-community” mode, sets of stocks actively avoiding one another), is the only candidate that is neither tautological nor part of the μ_1 family. Its case:

z -scored COVID AUC	0.775 (vs λ_2 ’s 0.759)
redundancy with λ_2 (z -scores)	+0.511 (against -0.915 for $\mu_1(\mathbf{B}_n)$)
redundancy with $\mu_1(\mathbf{B})$ (levels)	-0.668
independent test, reports/probe/ (6,315 days, 360 stocks, 17 pre-registered crises)	best modularity signal at AUC 0.760 vs λ_2 ’s 0.617 $Q(t)$, Paper 3’s own indicator, scores 0.538

The same quantity winning on two unrelated constructions is suggestive, and §7 now supplies a mechanism rather than a coincidence. **But it is a lead, not a result.** On FAAMUNG the difference is not significant ($\Delta\text{AUC} +0.015$ with CI $[-0.115, +0.148]$), and μ_N fails the same sliding-event null that λ_2 fails, at $p = 0.345$. This dataset supports the framing; it cannot support the claim.

Three possible directions

The three headings below follow from the results above. Everything after each heading is a suggestion rather than a recommendation, and the choice of which (if any) to pursue is yours.

D1. Validate λ_2 itself before comparing anything to it. §6 shows the incumbent indicator is not established on the data it was published on.

Suggestion. One way to settle this would be a testbed with many events scored under a single pre-registered detector, with a multiplicity correction. The apparatus for that already exists in `reports/probe/` (17 crises over 25 years, where λ_2 scores 0.617), so the cost would be small. It might be worth doing before the others, since it would establish whether the rest of the programme has a well-defined target at all, but that ordering is a judgement call.

D2. Characterise the departure from the normalised limit. §2 gives an exact statement of what **B** and **L** share, and §7 shows the remaining difference is carried entirely by the degree sequence, which is also where the detection improvement appears.

Suggestion. A natural object of study might therefore be the deviation itself: how far adding degrees moves **B** away from **B_n**, in which modes, and whether that displacement is the informative part. This looks to me like the version of the original question that does not collapse into an identity, and so perhaps the most promising candidate for a theory framework, though whether it is worth formalising is better judged by you.

D3. Test $\mu_N(\mathbf{B})$ properly, gated on D1.

Suggestion. This would probably only be worth attempting on a testbed with enough events to separate 0.775 from 0.759, which FAAMUNG is not. If it is attempted, I would suggest pre-registering both the direction and the detector before looking at the results, given how easily §6 shows this data produces apparent wins.

Reproducing this. `modularity_indicator/` contains `build_series.py` (re-runs Algorithm 2, computes the menu), `evaluate.py`, `robustness.py`, `significance.py`, `figures.py`, and `report_numbers.py`, which prints every number in this report. Run in order with `../.venv/bin/python`; total runtime under a minute. Nothing in `paper1_gmrf_laplacian/` is modified; it is read only. Companion documents: `reports/progress_report.tex` (running log) and `reports/2026-08-19_discrepancy_explained.tex` (the earlier three-paper exploration).